RSCTF TECHNICAL PRACTICE PAPER

RSCTF KING OF THE HILL HANDBOOK: BOOT2ROOT AND LEADERBOARD FORMATS

Dimas Maulana
RSCTF Project · Competition Platform

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Scoring policy: one constant formula per KotH format; no scoring versions

ABSTRACT

RSCTF defines two King of the Hill (KotH) formats with different competitive meanings. ***Boot2Root KotH*** *is an exclusive-control contest over one shared machine: teams acquire the hill, retain control, and accept responsibility for service health.* ***Leaderboard KotH*** *is a concurrent application or protocol contest: every eligible team may complete each challenge-native wave, and RSCTF compares every completed result with the best result from that same wave. Boot2Root uses the constant score 100R(0.25A + 0.55C + 0.20√AC). Leaderboard uses the constant wave score 100[0.95(S/S*)^(3/4) + 0.05K], where K is the unique Crown. An exact top-score tie gives every tied team full relative-performance credit and gives no team the Crown premium. Missing or incomplete teams receive zero for that wave. Points are not divided by field size, there is no separate winner or streak multiplier, and failed hacking attempts are not negative points. The managed arena reports bounded native evidence with a credential tied to its exact target lifecycle; RSCTF retains normalization and point authority. Authenticity, replay protection, exact runtime identity, independent checking, and challenge-level proof design determine whether evidence is admitted. Boot2Root capabilities rotate with scheduled pristine crown cycles. A Leaderboard arena remains persistent across rounds and epochs; its event token changes only through explicit security rotation, while a stopped runtime or repeated functional failure invokes health recovery. Both formats use scoped capabilities, bounded hill aggregation, and finalized epoch settlement. This paper is the canonical deployed scoring and operations contract.*

Keywords: *King of the Hill; Boot2Root; Leaderboard KotH; relative scoring; Crown; anti-cheat; crown cycle; RSCTF*

Document status: current implementation contract for the repository revision containing this paper. Public names are Boot2Root and Leaderboard; the stable internal claim-source identifiers remain `Marker` and `Api`. Those identifiers are not selectable scoring versions. This is a technical-practice report, not a claim of peer review or empirical validation.

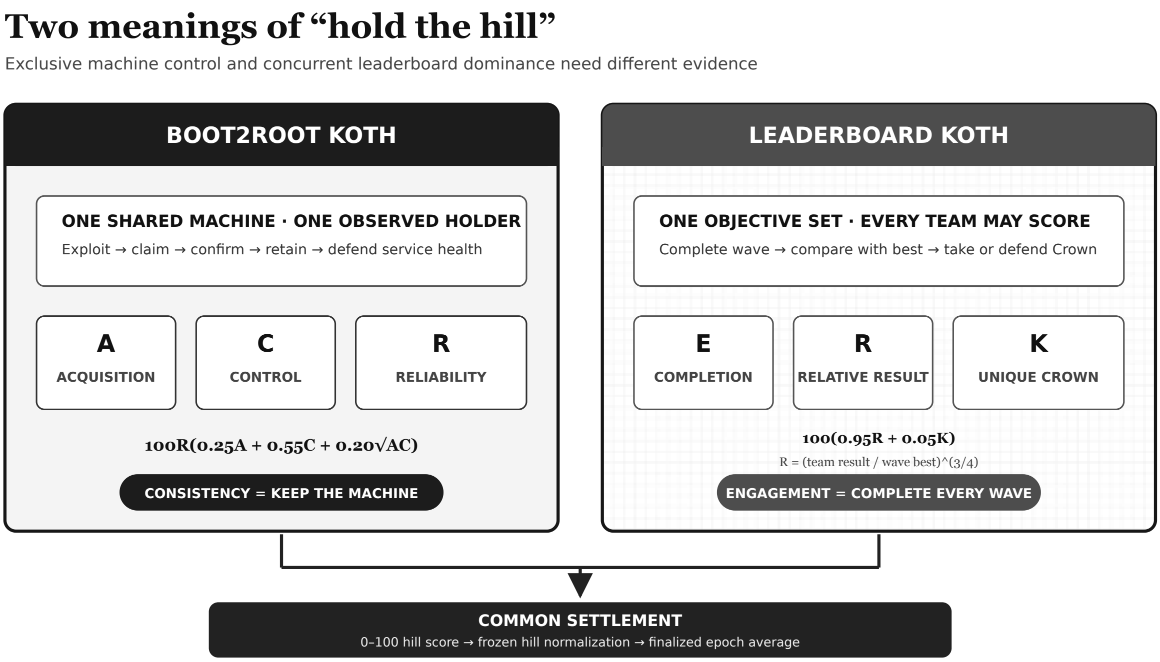

Figure 1. Boot2Root measures exclusive machine control. Leaderboard measures fresh, relative concurrent performance and one recurring per-wave Crown.

START HERE: KOTH IN 90 SECONDS

Choose the hill's format

Boot2Root means one shared machine and one holder:

1. exploit or administer the hill;
2. place the current per-hill capability in `/koth/king` ;
3. remain healthy through confirmation; and
4. retain control until takeover or the pristine reset.

Leaderboard means every eligible team may score concurrently:

1. use the challenge's player-facing mechanic;
2. finish a fresh, verified run in each wave;
3. maximize the published official result relative to the field; and
4. take or defend the unique Crown.

Players never call the managed evidence-reporting endpoints.

The two constant formulas

```
BOOT2ROOT
Core  = 0.25A + 0.55C + 0.20 * sqrt(A * C)
Local = 100 * R * Core

LEADERBOARD
R_t   = 0 if incomplete; otherwise (S_t / S*_t)^(3/4)
K_t   = 1 for the unique Crown; otherwise 0
Wave  = 100 * (0.95R_t + 0.05K_t)
```

The first formula uses acquisition `A` , control `C` , and reliability `R` . The second uses team result `S_t` , best result `S*_t` , relative performance `R_t` , and Crown indicator `K_t` for each finalized wave.

What resets and what settles

For **Boot2Root**, several ticks form a scheduled **crown cycle**. RSCTF then pauses the hill, destroys the old container, creates a pristine replacement, issues new cycle capabilities, proves readiness, and resumes scoring.

A **Leaderboard** arena has no scheduled Crown reset. It stays online across rounds and epochs. RSCTF checks it every round and replaces it only when the runtime stops or three consecutive challenge-attributed checks are unhealthy. Recovery time is field-wide void, and each team's event capability survives.

Several ticks form an **epoch**. **Projected** includes unfinished evidence; **Settled** includes finalized epochs and determines official rank.

1. SCOPE AND DESIGN REQUIREMENTS

1.1 Why the formats are separate

A Boot2Root hill asks who controls one shared target. Its natural evidence is exclusive identity, duration, acquisition, and service responsibility. A concurrent application may instead expose puzzles, transactions, simulations, or protocol objectives that many teams can complete at once. Treating the second as a one-holder marker creates last-write races; treating the first as an external point feed transfers scoring authority away from the platform.

RSCTF therefore freezes one competitive meaning per hill:

Table 1. Fixed public formats and internal claim-source identifiers.

Public format	Internal source	Concurrent scorers	Evidence
Boot2Root	`Marker`	at most one observed controller	acquisition, control, reliability
Leaderboard	`Api`	every eligible team	completion, relative performance, Crown

An API credential configured before official scoring selects internal source `Api` ; otherwise the hill uses `Marker` . The source cannot change after scoring begins. “Leaderboard” describes the competition, not its transport: the player surface may be HTTP, a binary protocol, a simulator, or a game server.

1.2 Shared invariants

Both formats preserve these requirements:

1. **Constant policy.** Each format has one formula and fixed coefficients.
2. **Bounded result.** A hill-epoch and aggregate epoch remain in `[0,100]` .
3. **Current identity.** Evidence belongs to the exact game, hill, round, lifecycle record, runtime attempt, target, and container. Boot2Root also binds the scheduled cycle capability; Leaderboard binds the current event-token generation.
4. **Independent health.** A functional checker, not the evidence reporter, decides whether the shared application works.
5. **No stale carry.** A missing team in a valid Leaderboard snapshot becomes zero; a missing field snapshot is void. Neither repeats old evidence.
6. **Infrastructure neutrality.** Provisioning, recovery, readiness, incomplete issuance, and attributable platform failures do not penalize participants.
7. **Auditable settlement.** Immutable wave evidence precedes final rollups, and retries cannot create duplicate credit.
8. **Frozen bounded state.** Rosters, hills, cadence, images, weights, objective identities, request sizes, replay memory, and retained evidence are bounded.

1.3 Relation to other systems

Bock, Hughey, and Levin describe an instructional KotH in which taking a shared machine also creates responsibility for critical services [2]. Boot2Root follows that exclusive-control interpretation while adding scheduled pristine replacement and fixed epoch settlement.

CTFd documents a KotH agent that reports a current identifier and awards a configured reward per check [3]. RSCTF additionally records provisional control, confirmation, acquisition, duration, and reliability.

rCTF documents an HMAC-signed dynamic-score endpoint accepting absolute `userId/points` setters [5]. RSCTF retains exact-body signing and replay resistance, but Leaderboard never accepts external points or team IDs. The platform normalizes bounded evidence, writes omitted-team zeros, detects leaders, and computes every score itself.

FAUST CTF is an Attack & Defense system rather than a shared hill, but its repeated rounds and service checks provide an operational comparison [4]. Its topology and formula are not RSCTF KotH.

2. COMPETITION PROTOCOL

2.1 Boot2Root protocol

RSCTF runs one managed shared container per Boot2Root hill. Every eligible team receives a high-entropy bearer capability scoped to the exact hill and reset attempt. A team claims by writing it to:

```
/koth/king
```

The checker reads the marker immediately before and after a functional probe. A stable eligible capability creates control and responsibility evidence. The first healthy observation is provisional; the configured consecutive healthy streak confirms acquisition. A different capability begins a new claim.

The previous cycle's team, or tied teams with the most confirmed healthy control, normally receives the configured opening cooldown. If blocking all tied leaders would leave no challenger, RSCTF omits the block. A blocked tick is removed from that team's personal opportunity denominator.

2.2 Leaderboard protocol

Each team receives one opaque, high-entropy capability for the game and Leaderboard hill. The player pastes only that value into the challenge. The challenge scopes it to its configured game and hill, exchanges it with RSCTF, and receives the authoritative team name plus a SHA-256 pseudonym. No local crew ID or crew name is accepted in event mode. The challenge discards the raw token and records only the lowercase pseudonym beside independently verified events. The platform-managed target keeps the lifecycle-bound reporter credential outside player responses and narrower child-process environments.

For every active scoring round, the target reporter fetches a context bound to the exact runtime, objective schema, eligible capability hashes, and a published settlement window. It filters its untrusted event ledger against that set and submits zero or more finalized waves. Each wave contains:

- a stable wave ID and server-confirmed end time;
- one completed `activity` ratio and one to sixteen objective ratios per team;
- an ordered `objectiveIds` list defining those positions; and
- one `isCrown` assertion for a unique positive leader, otherwise none.

The signed body contains `tokenHash`, not the bearer token. It contains no team ID, floating-point platform score, integrity multiplier, or point value. RSCTF resolves hashes against current capabilities and reports finalized-wave plus unique submitted/recognized-team counts.

Within one scoring context, every accepted finalized wave is append-only. A later snapshot may add a new wave but cannot alter or remove an earlier one.

The capability survives health recovery and runtime replacement. A player may explicitly rotate it after suspected exposure. Rotation immediately removes the old digest from the eligible set and clears only that team's current unsettled rows; every other team's evidence and already settled epoch evidence remain immutable. The player must reconnect with the replacement token.

The first accepted snapshot freezes the exact objective IDs and order. The schema survives credential rotation. A later reorder, rename, insertion, or deletion is rejected even when the component count stays the same.

Settlement runs 20 seconds behind the live round boundary. The checker waits for the published cutoff without holding a hill lock; the next round's window starts at the previous cutoff, so finalized-wave end times are covered without gaps. The checker then allows a bounded six-second arrival window, reads the complete snapshot, performs an independent probe, and reads it again. Only a byte-equivalent current snapshot plus a healthy verdict produces dense results for every frozen eligible team across every finalized wave. Omitted teams receive zero in that wave. A changing snapshot or unhealthy shared application voids the field-wide checker tick.

The final published window end is the Leaderboard scoring cutoff. A challenge must stop opening scoreable waves that cannot finish before it; the final 20-second reserve exists for the functional check and durable settlement, not late gameplay.

Leaderboard names one challenge-native Crown when a positive finalized wave has one unique leader. An exact top tie has no Crown. Leaderboard does not use the Boot2Root marker, provisional confirmation, or champion cooldown. Multiple teams can score in the same wave.

2.3 Source-aware runtime lifecycle

Boot2Root runs the durable sequence at every scheduled Crown boundary:

```
finalize → pause → audit/snapshot → destroy
        → create → issue → readiness → resume
```

Leaderboard uses that state machine for initial provisioning, event cleanup, and health recovery—not as a scoring clock. A stopped managed runtime starts recovery immediately.

Otherwise recovery requires three consecutive committed `Mumble` or `Offline` functional verdicts for the same exact runtime and attempt. `InternalError` does not count because uncertain platform inspection must not destroy a healthy arena. Recovery recreates the frozen image, clears transient snapshot/session state, preserves event capabilities, proves readiness, and then resumes. A retry continues the stored phase instead of starting a parallel replacement. Provisioning and recovery create no scoring opportunity.

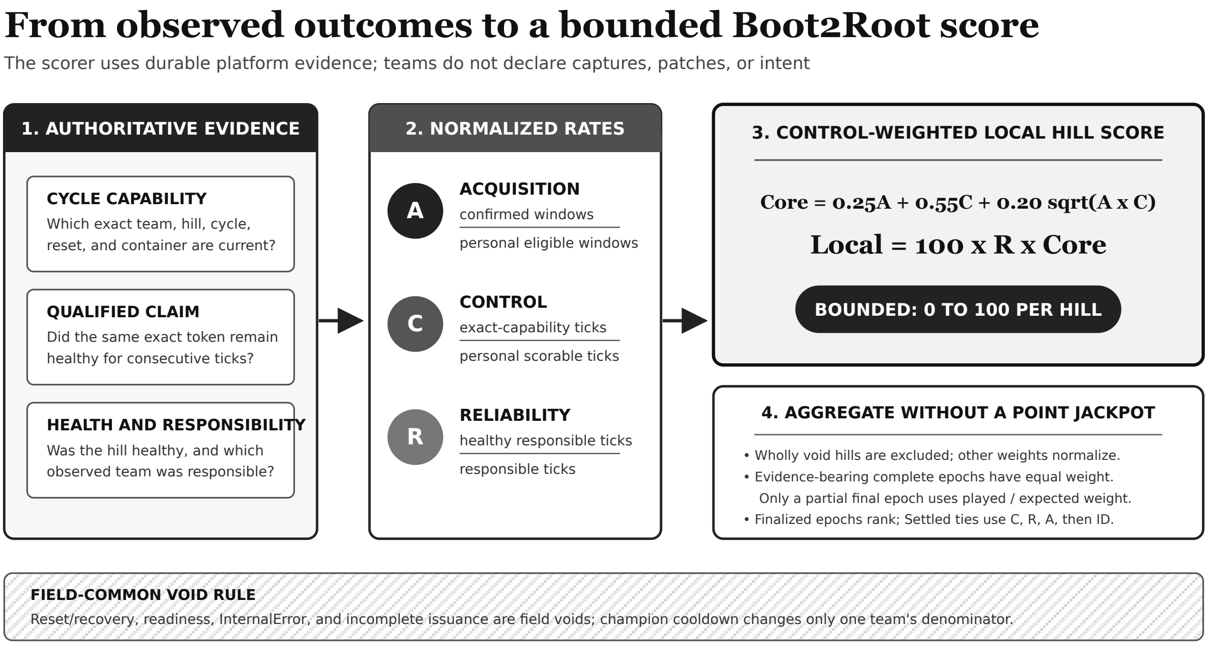

Figure 2. The Boot2Root branch shown here admits only current evidence; Leaderboard follows the same immutable-evidence boundary with per-wave relative metrics.

2.4 Exact attribution and voids

At most one immutable checker result exists per game, hill, and scoring round. Scorable evidence must match the frozen hill, active lifecycle record, runtime attempt, container, deadline, capability window, and—for Leaderboard—the objective schema. For Boot2Root, that lifecycle record is the scheduled crown cycle.

`InternalError`, incomplete capability issuance, and uncertain runtime inspection are platform voids. For Boot2Root, a responsible holder may receive the last authoritative availability verdict before recovery. For Leaderboard, a shared failure is field-wide void because it cannot be fairly assigned to one participant.

3. BOOT2ROOT SCORING

For team `i`, hill `h`, and epoch `e`, define:

Table 2. Boot2Root evidence counters.

Symbol	Evidence
x	crown-cycle windows in which the team confirmed acquisition
y	windows in which the team had an eligible opportunity
u	scorable ticks controlled by the exact capability
s	personally eligible ticks after void/cooldown removal
b	responsible ticks with checker verdict `Ok`
d	ticks for which the team was responsible

The three rates are:

$$A = x\ /\ y, \qquad C = u\ /\ s, \qquad R = b\ /\ d\ .$$

An empty denominator returns zero. The acquisition/control core is:

$$B^M = 0.25A + 0.55C + 0.20\sqrt{AC}\ .$$

The local hill score is:

$$H^M = 100RB^M\ .$$

Control has the largest direct coefficient. The geometric term rewards a team that both takes and retains the hill; reliability constrains the entire score.

3.1 Worked Boot2Root example

Suppose a team confirms one of two eligible windows, controls three of eight eligible ticks, and stays healthy in four of five responsible ticks:

$$A = 0.5, \qquad C = 0.375, \qquad R = 0.8\,.$$

$$B^M = 0.25(0.5) + 0.55(0.375) + 0.20\sqrt{0.5(0.375)}\,.$$

Thus `B^M = 0.41785` and the local score is `33.43`. Improving control raises the main term and the acquisition/control balance term; breaking the service reduces every earned control point through `R`.

4. LEADERBOARD SCORING

4.1 Fresh completion and native normalization

For team `i` and finalized wave `t`, activity is a completion gate. A scored row must report a fresh server-confirmed completion for that wave. Partial progress, polling, a previous result, or an omitted row produces zero.

For native objective `j`, every integer budget satisfies:

$$0 \le o^{+}_{itj} \le o^{*}_{itj} \le 10^{12}\,.$$

RSCTF normalizes each objective independently and takes their equal mean:

$$O_{it} = \frac{1}{m}\sum_{j=1}^{m}\frac{o^{+}_{itj}}{o^{*}_{itj}}\,.$$

A `9/10` result and a `900/1000` result both contribute `0.9`; a larger unit cannot become an accidental weight. A challenge such as Rythme may publish one combined official score as its single objective. The reporter still submits an integer ratio rather than platform points.

4.2 Relative performance without field dilution

Let `O*_t` be the highest positive completed objective result in wave `t`. Relative performance is zero without fresh completion. Otherwise:

$$R_{it} = \left(\frac{O_{it}}{O^{*}_{t}}\right)^{3/4}\,.$$

The three-quarter-power curve is fixed. It preserves order, maps the best result to one, and keeps close competitors close. It does not divide a point pool by the number of teams. Adding a new participant therefore cannot reduce an existing team's points unless that participant posts a new best result.

4.3 One Crown only for a unique leader

`K_it` is one only for the wave's unique Crown and zero for every other team. The Crown must have the highest positive completed result, and that result must belong to exactly one team. If two or more teams tie for the highest result, each receives full relative-performance credit and `K_it=0`; no timestamp, incumbent state, roster order, or stable identity breaks the scoring tie.

The Crown and unique first place are the same award. This prevents an arbitrary transport-order advantage while keeping the recurring five-point target for a team that strictly leads the wave. Every Crown holder must complete fresh play in that wave.

4.4 Constant wave score and epoch mean

The local result for one finalized wave is:

$$H^{L}_{it} = 100\left(0.95R_{it} + 0.05K_{it}\right)\,.$$

Performance supplies at most 95 points and the recurring Crown supplies five. There is no separate first-place bonus, participation bonus, exploit-status multiplier, or growing streak multiplier. The official challenge result may naturally include legal mechanics or an exploit-derived multiplier; RSCTF does not convert a binary “exploited” label into points.

For `W` finalized waves in an epoch, RSCTF gives every wave equal influence:

$$H^{L}_{i} = \frac{1}{W}\sum_{t=1}^{W} H^{L}_{it}\,.$$

Several waves may finish inside one RSCTF checker round. Their signed summary retains the wave count, so later epoch aggregation still weights each wave equally rather than each transport submission equally.

Table 3. One wave with native results 150, 20, 5, and 1.

Native result	Relative R	Crown K	Wave points
150	1.0000	1	100.00
20	0.2207	0	20.96
5	0.0780	0	7.41
1	0.0233	0	2.22

If several teams score near 150, each receives nearly 95 performance points; their scores do not collapse because the field is crowded. If only one team completes with a positive result—even a native result of `1`—it is best for that wave and receives 100. Teams tied at the best positive result each receive 95 because the wave has no unique Crown. Missing teams receive zero for the wave.

Failed hacking attempts are not subtracted. Invalid, forged, replayed, stale, or wrongly scoped evidence is rejected or excluded as telemetry. Section 7 defines the challenge-design obligations that make a completion meaningful.

5. HILL AGGREGATION, EPOCHS, AND RANK

Before scoring, each hill receives a frozen weight `w_h` in `[0.8,1.2]`. Let `z_he = 1` when hill `h` has field-wide scorable evidence in epoch `e`, and zero otherwise. The team epoch result is:

$$E_{ie} = \frac{\sum_h z_{he} w_h H_{ihe}}{\sum_h z_{he} w_h}.$$

A wholly void hill contributes no numerator or denominator. Once a Leaderboard hill has field evidence, an omitted team retains its explicit zero rather than removing the hill from its personal calculation.

A complete epoch has weight `q_e = 1`. If the event ends after `r` of `n` configured ticks in the final epoch, then `q_e = r/n`. The event score is:

$$T_i = \frac{\sum_e q_e E_{ie}}{\sum_e q_e}.$$

There is no late-epoch multiplier. Projected includes open epochs; Settled uses only finalized epochs.

Official KotH rank sorts by:

1. Settled score descending;
2. Control for Boot2Root or Objective rate for Leaderboard;
3. Reliability for Boot2Root or Crown-share rate for Leaderboard;
4. acquisition-window count or completed-wave count; and
5. stable participation ID.

The live projection never breaks an official tie. Displayed KotH ranks are ordinal; the final ID provides deterministic ordering.

Table 4. Source-aware scoreboard metrics.

Badge	First metric	Second metric	Third metric	Crown state
Boot2Root	Acquisition	Control	Reliability	confirmed / pending
Leaderboard	Completion	Relative objective	Crown share	per-wave Crown

6. FAULT AND ADMISSION POLICY

Table 5. Evidence admission and fault treatment.

Condition	Boot2Root	Leaderboard
Missing own evidence in valid tick	no control/health credit	explicit zero row
Missing or changing whole snapshot	not applicable to marker feed	field-wide void
Functional checker fails globally	attributed only when responsibility is authoritative; otherwise void/recovery	field-wide void
Invalid/revoked capability	no claim	row rejected or unrecognized
Old lifecycle/runtime/container	no claim	signed context rejected; event token remains usable after recovery
Replay, old timestamp, or future wave	not a marker operation	request rejected
Objective IDs/order changed	not applicable	request rejected
Provisioning/recovery/readiness/incomplete issuance	void	void
Failed player exploit attempt	no score by itself	no negative points; telemetry/rate limits still apply

A `200` response from the signed endpoint only stages a snapshot. It does not award points. The checker still has to bracket the snapshot around a healthy probe and persist immutable evidence.

7. SECURITY AND ANTI-CHEAT CONTRACT

7.1 HMAC authenticates origin, not truth

RSCTF injects a lifecycle-bound HMAC credential into the exact managed target that owns native gameplay truth. A valid signature shows that this target produced an exact body; it does not show that a compromised target measured honestly. The credential cannot select points, cross into another hill, or survive a target reset.

Keep the credential out of handouts, player responses, browsers, logs, repositories, backups, and child gameplay processes. Reporter code should read an immutable in-memory snapshot or a loopback-only feed and call only the private rsctf origin. PostgreSQL backups contain symmetric credentials and remain sensitive. A transport gateway receives neither the credential nor scoring state.

7.2 Make score require play

A defensible Leaderboard challenge:

1. counts completed challenge-relevant work, not requests, packets, page views, or connection volume;
2. issues unpredictable, expiring, one-use tasks or proofs;
3. binds each task and result to the capability hash that began it;
4. verifies results server-side and makes replay idempotent;
5. publishes fixed completion rules, objective IDs/order, and denominators;
6. bounds sessions, requests, evidence retention, pagination, and queue work;
7. keys team quotas by capability identity rather than source IP;
8. isolates per-team work so one team cannot exhaust shared admission;
9. exposes an ordered feed cursor and fails closed on a retention gap; and
10. keeps the functional checker independent and read-only.

Fuzzing, failed exploit development, and malformed challenge traffic are part of hacking unless event rules prohibit a specific action. They do not deserve an automatic score penalty. Rate limits and bounded work protect availability; only verified completed outcomes create positive scoring evidence.

7.3 Capability and context scope

RSCTF resolves a submitted `tokenHash` only against the current event capability for the exact game, hill, and official participation. The signed context independently binds the round, target, lifecycle record, runtime attempt, container, reporting-configuration revision, and frozen objective-schema hash. A reporting credential change therefore invalidates the previous deduplication fence. Compatibility fields remain named `cycleNumber`, `resetAttempt`, and `cycleEndsAt`; for Leaderboard, they fence runtime recovery and the event scoring cutoff rather than a scheduled Crown reset. Raw bearer tokens enter only the narrowly scoped authentication exchange and never enter the signed reporter request. Unknown and rotated hashes cannot become arbitrary team identities.

The wire path remains `/api/v1/...` for compatibility. `v1`, `Marker`, and `Api` are protocol or storage identifiers, not scoring-policy versions. The database stores no selectable KotH formula column.

8. ORGANIZER OPERATIONS

8.1 Preflight

Before official scoring:

- choose and publish each hill's format, cadence, bounded weight, checker contract, and appeal policy;
- confirm at least two accepted teams receive distinct current capabilities;
- for Boot2Root, run a pristine replacement and reject every previous cycle capability afterward;
- verify the checker before and after evidence reads;
- for Boot2Root, verify provisional confirmation, takeover, responsibility, and enforceable cooldown;
- for Leaderboard, verify the arena stays on the same runtime through an ordinary Crown boundary, the same event token works after a forced health recovery, explicit rotation rejects the prior token and only that team's unsettled rows, and then freeze the final objective IDs/order and verify HMAC, submitted-wave and recognized/submitted equality, independent normalization, exact-tie no-Crown behavior, explicit zero, reporter restart, stale hash, replay, clock skew, feed gap, and changing-snapshot behavior; and
- run a complete accelerated epoch at expected peak capacity, including one deliberately overloaded team, while confirming that other teams and the reporter retain reserved capacity.

The official snapshot records roster, hills, images, source, cadence, and weights. Objective identity freezes on the first accepted Leaderboard snapshot. These score-affecting inputs cannot change after the competition boundary.

8.2 During play

Monitor checker gaps, void reasons, Boot2Root reset phases, Leaderboard health recovery, runtime identity, snapshot freshness, recognized-team counts, cursor lag, queue saturation, and reporter errors. Pause scoring before forcing a managed target reset; verify that the stale lifecycle credential is rejected, then accept fresh evidence from the replacement before resuming.

At event end, reject late evidence, complete authoritative in-flight checks within the deadline fence, settle the proportional tail, and publish awards from Settled results. Retain official configuration, capability issuance and revocation, immutable checker rows, Leaderboard wave evidence, lifecycle receipts, and epoch rollups through the appeal period.

8.3 Signed body summary

The complete wire contract and signing example are in the organizer reporting guide. The core body is:

```json
{
  "context": "<64 lowercase hex characters>",
  "objectiveIds": ["official-score"],
  "waves": [
    {
      "waveId": "heat-42",
      "endedAtUnixMs": 1786200000000,
      "teams": [
        {
          "tokenHash": "<sha256 of current capability>",
          "activity": {
            "earned": 1,
            "possible": 1
          },
          "objectives": [
            {
              "earned": 150,
              "possible": 150
            }
          ],
          "isCrown": true
        }
      ]
    }
  ]
}
```

9. VALIDATION AND LIMITATIONS

The equations prove boundedness and zero conditions. Unit, database, and lifecycle tests establish behavior for exercised cases. Neither proves that an organizer chose competitively valid objectives.

Organizers should measure completion and objective distributions, Crown concentration, missing/void frequency, reporter lag, feed-retention margin, marker confirmation failures, control changes, repeat winners, Boot2Root reset latency, Leaderboard recovery frequency, and score sensitivity to targets and cadence.

A randomized checker sample cannot prove continuous state between observations. Capability hashing cannot repair low-entropy tokens. HMAC cannot prove arena correctness. Equal normalization cannot prove equal objective difficulty. The formula cannot detect off-protocol collusion. A field-wide outage is void, so weak resource isolation could let one participant manufacture a veto. These remain challenge-design and operational responsibilities.

10. CONCLUSION

RSCTF does not treat concurrent KotH as Boot2Root with a different transport. Boot2Root rewards qualified acquisition, sustained exclusive control, and service reliability. Leaderboard rewards fresh performance relative to the same wave's field and adds one recurring five-point Crown premium.

Both formulas are constant and platform-owned. Dense zero rows prevent stale team scores; field-wide voids prevent infrastructure failures becoming participant penalties. Exact identity fences, independent checking, source-aware runtime lifecycle, bounded hill normalization, and finalized epochs make results reproducible. Fairness still depends on published rules, checker coverage, objective design, reporter/child-process isolation, network equality, and incident review.

APPENDIX A. FREQUENTLY ASKED QUESTIONS

A.1 Does Leaderboard read `/koth/king`?

No. Boot2Root has one holder and acquisition/control/reliability evidence. Leaderboard has concurrent teams and completion/relative-performance/Crown evidence. Only the lifecycle infrastructure is shared.

A.2 Can the target reporter submit points or team IDs?

No. It submits integer evidence ratios, an optional per-wave Crown assertion, and current capability hashes. RSCTF resolves identity, normalizes relative performance, validates the Crown, and computes points.

A.3 Why the three-quarter-power curve?

It preserves rank and maps the best completed result to one while keeping close results competitive. It also avoids a shared point pool: field size alone cannot dilute an existing score.

A.4 Why normalize objectives separately?

Raw sums let the largest numeric unit dominate. Independent ratios give every objective the same `[0,1]` range before the equal mean.

A.5 What if my team is omitted from a valid snapshot?

RSCTF writes an explicit zero for that wave. It never repeats an old result.

A.6 What if the whole snapshot or application is unavailable?

The checker tick is field-wide void and enters no team's denominator.

A.7 Do failed hacking attempts lose points?

No. Only meaningful verified outcomes add evidence. Invalid traffic can be rate-limited and audited, but it is not an integrity score multiplier.

A.8 Does runtime replacement erase settled evidence?

No. A scheduled Boot2Root reset removes transient containers, sessions, claims, and cycle capabilities. Leaderboard health recovery removes transient runtime and snapshot state but preserves event tokens. Immutable results and finalized rollups remain in both formats.

A.9 Is there formula versioning?

No. Boot2Root and Leaderboard are distinct formats with one constant formula each. Protocol path `v1` is unrelated to scoring policy.

A.10 Can automation participate?

Yes, subject to event rules. RSCTF validates evidence and scope rather than guessing whether a participant is human or automated.

APPENDIX B. NOMENCLATURE

Table 6. KotH nomenclature.

Symbol or term	Definition	Symbol or term	Definition
A, C, R	Boot2Root acquisition, control, and reliability rates	z_{he}	field-evidence switch for one hill and epoch
S_{it}, S^*_t	team and best completed native results in wave `t`	q_e	complete or proportional final-epoch weight
R_{it}	three-quarter-power relative performance in wave `t`	**Settled**	official value using finalized epochs
K_{it}	unique Crown indicator in wave `t`	**Projected**	information that also includes open evidence
H^M, H^L	local Boot2Root and Leaderboard scores in `[0,100]`	**Field void**	sample excluded from every team's denominator
w_h	frozen hill weight in `[0.8,1.2]`	**Explicit zero**	omitted or incomplete Leaderboard team in a valid wave

APPENDIX C. IMPLEMENTATION TRACEABILITY

Paths are relative to the repository revision containing this paper.

Table 7. Implementation traceability.

Responsibility	Source of truth
Official source, roster, hill, cadence, image, and weight snapshot	`src/services/ad/engine/koth_cycle/config.rs`
Source-aware scheduled reset and health-recovery lifecycle	`src/services/ad/engine/koth_cycle/lifecycle/`
Lifecycle-bound target credential creation and injection	`src/services/ad/koth_reporter.rs` , `src/migrations/m0112_koth_target_reporters.rs`
Boot2Root claim and acquisition	`src/services/ad/engine/koth_cycle/claims.rs`
Exact marker read	`src/services/ad/engine/koth_marker.rs`
Leaderboard event-token authentication and rotation	`src/services/ad/koth_api_capability.rs` , `src/controllers/game/koth/tokens.rs`
Signed context, HMAC, replay, schema, and submission	`src/controllers/game/koth/api/` , `api_contract.rs`
Stable finalized-wave snapshot read and relative curve	`src/services/ad/engine/koth_api.rs`
Checker persistence, dense zeros, and Crown validation	`src/services/ad/engine/checker/koth_api.rs`
Constant pure formulas	`src/controllers/game/koth/scoring_formula.rs`
Equal-wave SQL epoch aggregation	`src/controllers/game/koth/scoring/evidence.rs`
Final rollups	`src/controllers/game/koth/scoring/rollup/`
Board labels and rank	`src/controllers/game/koth/board.rs` , `web/src/components/KothScoreboardTable.tsx`
Constant Leaderboard schema and removal of formula selectors	`src/migrations/m0085_constant_leaderboard_scoring.rs`

Responsibility	Source of truth
Event-scoped Leaderboard token schema and live-token preservation	`src/migrations/m0086_koth_api_event_tokens.rs`
Finalized-wave contract and constant 95/5 relative scoring	`src/migrations/m0088_koth_api_wave_scoring.rs`

C.1 HTTP surface

Table 8. Player, operator, and managed-reporter HTTP surface.

Method and route	Purpose
`GET /api/game/{id}/ad/koth/{challengeId}/token`	caller's exact current-hill capability
`POST /api/game/{id}/ad/koth/{challengeId}/token`	explicitly rotate a Leaderboard event capability
`POST /api/v1/koth/capability/authenticate`	exchange one scoped event capability for the authoritative arena identity
`GET /api/game/{id}/ad/koth/{challengeId}/state`	lifecycle and Boot2Root holder state
`GET /api/game/{id}/ad/koth/scoreboard`	source-aware metrics, Projected, Settled, rank
`GET /api/game/{id}/ad/koth/timeline`	finalized/projected cumulative history
`GET /api/edit/games/{id}/ad/koth/state`	operator lifecycle and evidence view
`POST /api/edit/games/{id}/ad/koth/{challengeId}/recover`	idempotent lifecycle recovery
`GET/POST/DELETE /api/edit/games/{id}/ad/koth/{challengeId}/observer`	inspect or enable Leaderboard reporting; manage the legacy external credential
`GET /api/v1/koth/games/{id}/challenges/{challengeId}/context`	exact runtime/schema fence, event cutoff, and eligible hashes
`POST /api/v1/koth/games/{id}/challenges/{challengeId}/observations`	signed bounded evidence; never points

Wire DTOs use camelCase and Unix-millisecond timestamps.

C.2 Verification scope

The suite covers formula bounds and zero conditions, objective normalization, ordered schema identity, malformed evidence, HMAC scope, clock skew, replay, context rotation, current capability resolution, unknown hashes, dense zeros, snapshot bracketing, relative performance, exact-tie no-Crown handling, marker confirmation, API persistence past a scheduled Crown boundary, three-failure health recovery, cooldown denominators, tied champions, partial epochs, rollups, and ordinal ranks. The bundled challenge exercises one-use proofs, invalid traffic, capability hashing, bounded pagination, reporter restart, feed-gap failure, and exact signed bodies. Lifecycle and fixed-load results are recorded in `tests/load/REPORT.md`; they do not establish competitive fairness.

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